MLK V3.5 — Metrics by Class

Every number below was measured against the live API. Cold starts shown separately, never hidden in averages.

CLASS 1 — Health Check · GET /api/healthz

API liveness only — no audio, no kernel. Baseline server responsiveness.

1.7 ms
median (warm)

RUN	RESULT	STATE
1-10 (10 runs)	1.6 ms	warm
fastest	0.9 ms	warm
slowest	10.0 ms	warm

WARM-RUN STATISTICS	
Best	**0.9 ms**
Median	**1.7 ms**
Mean	**2.5 ms**
Worst (warm)	**10.0 ms**

Sub-2 ms typical

Server-side time. Browser round trips add network/proxy latency on top (~100-300 ms over the public internet).

CLASS 2 — Watermark Verify · POST /api/v1/verify

Cryptographic ownership check on an uploaded 5.3 MB WAV — includes the file upload.

22.8 ms
median (warm)

RUN	RESULT	STATE
run 1	22.8 ms	warm
run 2	15.0 ms	warm
run 3	32.4 ms	warm

WARM-RUN STATISTICS	
Best	**15.0 ms**
Median	**22.8 ms**
Mean	**23.4 ms**
Worst (warm)	**32.4 ms**

Under 35 ms, every run

Competitors do not offer in-audio cryptographic verification at any speed.

CLASS 3 — Full Master, 30-second track · POST /api/v1/ingest

Complete MLK V3.5 mastering pipeline on a 30 s stereo WAV (5.3 MB), mastered file returned.

1.35 s
median (warm)

RUN	RESULT	STATE
run 1	14.10 s	cold start
run 2	1.37 s	warm
run 3	1.34 s	warm

WARM-RUN STATISTICS	
Best	**1.34 s**
Median	**1.35 s**
Mean	**1.35 s**
Worst (warm)	**1.37 s**

~22x realtime

Cold start = first request after idle (Python/Numba warm-up). Keep-warm removes it in production.

MLK V3.5 — Metrics by Class (continued)

Full-length song benchmark plus how each class maps to what the industry offers.

CLASS 4 — Full Master, 3.5-minute song · POST /api/v1/ingest

Complete MLK V3.5 mastering pipeline on a 3 min 30 s stereo WAV (37 MB), mastered file returned.

8.72 s
median (warm)

RUN	RESULT	STATE
run 1	29.62 s	cold start
run 2	12.84 s	warm
run 3	8.72 s	warm
run 4	9.02 s	warm
run 5	8.51 s	warm
run 6	8.44 s	warm

WARM-RUN STATISTICS	
Best	**8.44 s**
Median	**8.72 s**
Mean	**9.51 s**
Worst (warm)	**12.84 s**

~24x realtime

Run 2 partially warm (workspace under load). Steady state: 8.4-9.0 s. Times exclude the customer's internet upload.

How each class compares to the industry

Metric class	MLK V3.5 (measured)	Industry typical	Advantage
Health check	**1-2 ms**	10-100 ms (typical API)	**Instant**
Ownership verify	**15-32 ms**	Not offered	**Exclusive**
Master 30 s track	**1.35 s**	1-3 min (AI platforms)	**>40x faster**
Master 3.5 min song	**8.5 s**	1-5 min (AI) / 1-5 days (human)	**>7-35x faster**

METHODOLOGY

All MLK V3.5 timings measured August 5, 2026 with curl against the live GravelKing Pro API (local server, near-zero network time).
Test material: generated stereo WAV fixtures (30 s / 5.3 MB and 3 min 30 s / 37 MB, 44.1 kHz 16-bit). Every request returned HTTP 200 with the mastered fil
Cold-start runs are labeled and excluded from warm statistics. Industry figures are typical published per-track turnaround times including their queue and rend
Realtime multiplier = audio duration / processing time (210 s / 8.6 s median = ~24x).